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EFFECT OF HEARING AIDS NOISE REDUCTION ON LISTENING EFFORT DURING CONTINUOUS SPEECH: CO-REGISTRATION OF PUPILLOMETRY AND EEG.

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Abstract

Concept of listening effort is critical for understanding challenges that hearing-aid users face in daily acoustic challenges situations. Understanding the impact of signal processing on listening effort is crucial for hearing-aid manufacturers. Among methods, pupillometry and electroencephalography (EEG) measures are the most cited due to their ability to be used during standard clinical speech perception assessments. In the present study, listening effort was assessed for 36 hearing-aid users. Two hearing-aid models were tested, with and without Noise Reduction (NR). The speech material consisted of 30-second continuous streams that were presented from loudspeakers to the right and left side of the listener (±30° azimuth) in the presence of 4-talker background noise (+180° azimuth). Participants were instructed to attend either to the right or left speaker and ignore the other in a randomized order. To maintain attention, a three-choice question followed regarding the content of the attended target audio clip. Behavioral results indicate an improvement of correct responses when NRs were activated. Pupillometry analysis indicates less pupil dilation with NRs, suggesting less listening effort. Alpha and theta brain waves analysis have not permitted to extract differences among conditions. This study confirmed that the use of NRs provides an improvement of comprehension and reduced listening effort.

Résumé

Les concepts de demande cognitive et d'effort d'écoute apparaissent aujourd'hui comme cruciaux dans la compréhension des défis auxquels se confrontent les porteurs d'aides auditives dans les situations du quotidien. L'évaluation de l'impact des traitements de signaux – tel les réducteurs de bruit – sur l'effort d'écoute apparait comme un enjeu majeur pour les fabricants d'aides auditives. Parmi les méthodes d'évaluation, les mesures de pupillométrie et d'électroencéphalographie (EEG) ont été exploités a de nombreuses reprises pour témoigner des changements du système nerveux reflétant l'effort d'écoute. Dans cette étude, l'effort d'écoute chez 36 utilisateurs d'aides auditives a été évalué. Deux modèles d'aide auditives ont été étudiés, avec et sans réducteurs de bruit. Au cours du test, il était demandé au sujet d'écouter un sujet d'actualité de 30 secondes tout en ignorant un sujet d'actualité concurrent. A ces deux sources, diffusées chacune a plus ou moins 30 degrés face au sujet, était ajouté un babble-noise diffusé à l'arrière du sujet. Le rapport signal sur bruit (SNR) résultant était de 3 dB entre le sujet d'actualité cible et les autres sources. A l'issue des 30 secondes, le sujet était soumis à une question a 3 choix forcés. Les résultats indiquent une amélioration des réponses des participants avec réducteurs de bruit activé. La dilatation pupillaire est moins importante avec réducteurs de bruit, suggérant une diminution de l'effort d'écoute. L'exploitation des données EEG (ondes alpha et thêta) n'a pas permis de mettre en exergue un effet des réducteurs de bruit. Cette étude confirme que l'utilisation de réducteurs de bruit permet une amélioration de la compréhension chez les porteurs d'aide auditive ainsi qu'une diminution de l'effort d'écoute souligne par l'analyse pupillometrique.

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Contents

Introduction

Acoustic challenging situations can arise in non-ideal listening environments caused by signal degradation. This degradation can emerge from a noisy background or from hearing loss, characterized by elevated hearing thresholds and reduced spectrotemporal resolution (McCoy et al., 2005; Shinn-Cunningham and Best, 2008; Pichora-Fuller et al., 2016). In such conditions, the individual expends extra effort to achieve successful processing, comprehension and remembering speech (McGarrigle et al., 2014). People with hearing-loss are particularly affected by signal degradation and require much more processing effort to identify sounds compared to those with normal hearing (Downs, 1982; Arlinger, 2003; Rönnberg et al. 2013). This concept of effort in listening situation is called listening effort and has recently been defined as "*the deliberate allocation of mental resources to overcome obstacles to goal pursuit when carrying out a listening task*" (Pichora-Fuller et al. 2016). The consequences of an increased listening effort can be mental fatigue leading to stress or increased days of sick leave (Gatehouse and Gordon, 1990; Edwards, 2007; Hornsby, 2013).

Concerning Hearing Aid (HA) development, the central topic is the improvement of signal processing algorithms in noisy environments since most of the HA users report a lack of benefit in environmental noise. Numerous studies report a benefit of HA signal processing algorithms on listening effort (Downs 1982; Gatehouse and Gordon 1990; Strauss et al., 2009; Sarampalis et al. 2009; Hornsby 2013; Ahlstrom et al. 2013).

Different types of methods and tools have been used to explore listening effort. These methods and tools are subjective rating questionnaires or scales (Humes, 1999, Gatehouse and Noble, 2004; Ahlstrom et al., 2013), dual tasks (performance measure on one task while the difficulty on a concurrent task varies) and physiologic measures such as changes in brain oscillations, pupil dilation, skin conductance and stress cues. Among these objective assessments of listening effort, pupillometry and electroencephalography (EEG) are the most cited due to their ability to be used during standard clinical speech perception assessments.

General findings on pupillometry have shown that pupil size was sensitive to changes in cognitive resources allocation. In general, pupil size increases with increasing task demands (Granholm, Asarnow, Sarkin, & Dykes, 1996). It has been shown that speech stimuli manipulations affect pupil dilation. For instance, pupil size tends to be larger when performance levels are low (Koelewijn et al., 2012; Krameret al., 1997; Zekveld et al., 2014). Among other, some studies suggest that the degradation of a speech stimulus increases pupil size (Winn et al., 2015). Wendt et al. (2017) have explored the effect of different NR schemes on pupil dilation using sentences from the Danish version of the Hearing In Noise Test (HINT). Results suggest a benefit of NR schemes on processing effort, illustrated by a decrease of peak pupil dilation (PPD) when NR schemes are used (Wendt et al., 2017).

Research using EEG has shown that neural oscillations exhibit changes associated with a wide variety of cognitive processes, particularly for alpha band (8-12 Hz) in parietal lobe and theta band (4-8 Hz) in frontal lobe. The theta activity in the frontal lobe, has been mostly linked to non-speech processing workload such as pitch discrimination (Wisniewski et al., 2018). Alpha parietal oscillations have been related to speech processing (Obleser and Weisz, 2012), and more precisely to attentional processes in active versus passive listening (Dimitrijevic et al., 2017) or different selective attention conditions (Dimitrijevic et al., 2019). However, studies has shown different outcomes : in some studies, alpha activity increases with more demanding tasks (Obleser et al., 2012; McMahon et al., 2016; Wisniewski et al., 2017a, Miles et al., 2017) whereas in others, alpha suppression has been marked as a sign of listening effort (Cartocci et al., 2016; Marsella et al., 2017; Miles et al., 2017; Hjortkjaer et al., 2018). Some authors suggest that these contradictory results are closely related to the linguistic complexity, the duration of the speech material and the background noise used (Peelle, 2012). In other words, when linguistic complexity or the duration increases, the signal of interest influences differently the alpha network.

The aim of the present study was to evaluate the effect of NR schemes on performances and listening effort of two commercially HAs. Performances were analyzed by studying correct responses. Listening effort was investigated by measuring changes in pupil dilation and brain oscillations during a listening task.

We hypothesized: (**H1**) By applying a NR scheme, participant performances should be improved; (**H2**) A benefit of the NR scheme on listening effort should be reflected on pupillometry measurement, with a reduced mean pupil dilation (MPD) with NR scheme and (**H3**) EEG measurements should reflect listening effort, characterized by variations of power alpha and theta brain waves with NR scheme.

Material and method

Participants

36 native Danish-Speaking adults with an average age of 64.2 ± 13.3 years participated in the study and signed a written consent form prior to study onset. All test participants were experienced hearing aid users with symmetrical, mild, sensorineural hearing loss. The pure-tone average of air conduction thresholds at 0.5, 1, 2 and 4 kHz (PTA4) was 47.5 ± 11.7 HL (see **Fig. 1**.).

Apparatus

The experiment was set up in a double-walled sound-proof booth. The experimental setup consisted of three loudspeakers positioned at ±30° and +180° azimuth relative to the participants. The loudspeakers in the front were the target speakers, and the loudspeaker in the back played a 4-talker babble noise. The eye tracker and a computer screen for displaying the instructions and the questions were positioned in front of the participants in a way not to cause acoustic shadowing. The spatial setup of the test is presented in **Fig. 2A.**

Stimuli were routed through a sound card (RME Hammerfall DSB multiface II, Audio AG, Haimhausen, Germany) and were played via loudspeakers Genelec 8040A (Genelec Oy, Iisalmi, Finland). Pupillometry and EEG devices were used to collect the physiological data. Pupil diameters of the left and right eyes were recorded by an SMI iView (SensoMotoric Instruments, Teltow, Germany), RED250 mobile system with a sampling frequency of 60 Hz. EEG data were recorded by a BioSemi ActiveTwo amplifier system (Biosemi, Netherlands) with a standard cap including 64 surface electrodes mounted according to the international 10-20 system with a sampling frequency of 1024 Hz. The cap included DRL and CMS electrodes as references for all other recording electrodes. All electrodes were mounted by applying conductive gel to obtain stable and below 50 mV offset voltage.

Speech Materials and Background Noise

The speech streams consisted of 30 seconds of Danish non-dramatic news clips of neutral content read by a male and a female. The attended speaker was randomized between each trial. Speech streams were preceded by 5 seconds of masker. This masker was a 4-talker babble noise, provided by Danish audiobooks. The sound pressure level was at 50 dB for the babble noise, 65 dB for the target and 62 dB for the distractor to generate a SNR of +3 dB. The timeline of one trial is presented in **Fig 2C.**.

Hearing Aids, Noise Reduction and Gender

Two HAs (HA1 and HA2) were tested during the experiment. To evaluate the impact of NRs, each HA were tested with and without NR (ON and OFF). As noted above, two-different speaker of different gender (Male and Female) were used as speech stimuli.

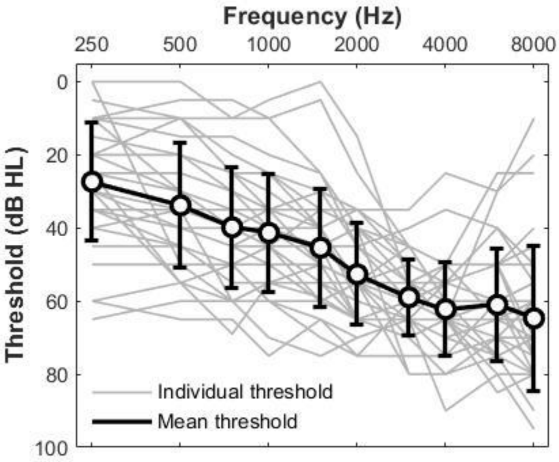

Figure 1. Individuals and average pure-tone audiograms of participants. Each grey line represents one audiogram participant. Black line represents average and associated standard deviation of all participants.

Procedure

There were 4 blocks of 20 trials, one per condition. Blocks were randomly distributed. The target and distractor were presented 5 seconds after the babble noise onset, and then continued for 30 seconds, followed by a three-choice question regarding the content of the attended target audio clip.

For each participant, the experiment started with 6 training trials. Before each trial, the participants were instructed on the screen to pay attention to the target on the right or the left side and ignore the talker on the other side and the babble noise. The location and the gender of the target speech was also randomized between each trial.

Behavioral measurement

To maintain the attention of the participant, each trial was followed by a three-choice question. The percentage of correct response was extracted for each condition. Only subject with more than 60% were kept for behavioral analysis. Among 36 subjects, 33 were kept for further analysis.

Pupillometry measurement

To analyze the pupillometry data, eye blinks were first detected as pupil diameter data with values two standard deviations (SD) below the trace's mean value and then removed. Missing gaps caused by blink removal were linearly interpolated 80ms before and 150 ms after the blinks to match the rest of the trace. Other high frequency artifacts potentially caused by unrelated physiological processes were also removed from the signal by means of a moving average filter with a symmetric rectangular window of 600 ms length.

Three criterions were defined to keep subject for further analysis: only trials with more than 60% reliable data were kept for further analysis and subject kept had more than 60% valid trials. Furthermore, only subject with more 60% good answers were kept in pupil analysis. Considering these criterions, 22 subjects were kept for further analysis.

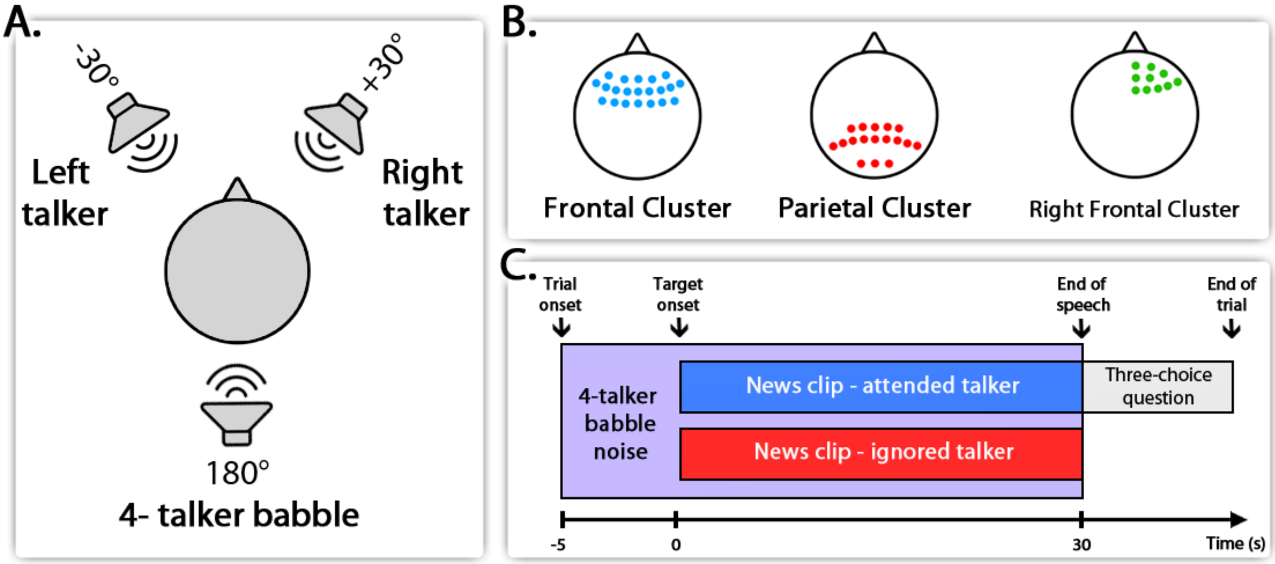

Figure 2. Experiment design A. Spatial setup of the experiment. **B.** Electrode clusters averaged in EEG results. **C.** Timeline of each trial.

As a normalization method, subtraction of the pupil baseline value (-4 to 0 sec., with 0 indicating the onset of the target) was used to extract task-related pupil activity. Data were averaged across the tested conditions (HA1 with NR OFF, HA1 with NR ON, HA2 with NR OFF and HA2 with NR ON). In order to investigate a potential effect of time, each data point within 5-second intervals was averaged together (*e.g.*, 0-5 sec., 5-10 sec. and so on). This also provided an opportunity to compare the results of pupillometry with EEG power spectral analysis in the same time intervals. MPD was applied since it is more robust compared to PPD in longer stimuli designs, as MPD extracts all the information within 30 seconds of data. In contrary, PPD usually happens only in the first few seconds of the target onset and gives no further information for the rest of the stimuli.

EEG measurement

The EEG trials were segmented from -5 to 32 seconds after stimuli onset (the last 2 seconds only included to avoid edge effects of the spectral perturbation). First, 50 Hz power line noise was rejected with a notch filter with quality factor of 25. Then, a 3rd-order zero-phase Butterworth bandpass filter with a cutoff frequency of 1 to 40 Hz was applied to the data, which was afterwards down-sampled to 256 Hz. Bad channels were automatically detected if they had values higher than three SD in more than 25% of the whole recording. A maximum of 3 out of 64 channels were detected as bad across all participants, in which case, the data were interpolated using spline interpolation in the EEGLAB toolbox (Delorme and Makeig, 2004).

To remove artifacts in EEG data, joint decorrelation (De Cheveigné and Parra, 2014), which is an improved method over denoising source separation (DSS), was applied. The bias filter in this method for denoising was chosen as the average of trials. Each of the extracted components were ranked according to the power of their mean divided by the total power, which implies that the first component has the strongest possible mean effect relative to overall variability and hence has the highest chance to be neural activity related. To decide how many of the components should be kept and then backpropagated to the sensor level, a surrogate procedure took place. If the score of the component was higher than the 95% confidence interval of the surrogate data, the component was regarded as neural activity; otherwise, it was discarded (De Cheveigné and Parra, 2014). It should also be noted that no trial was discarded due to poor signal quality and the results are based on the average of all recorded trials.

To assess how the EEG power spectra changed compared to the baseline, the event-related spectral perturbation (ERSP) method was used. The main characteristic of this method is that the EEG power over time within a predefined frequency band is displayed relative to the power of the same EEG derivations recorded during the baseline period (Makeig, 1993). The formula to calculate the ERSP is as follows:

$$ERSP_t\ (\%) = \frac{A_t - R}{R} \times 100$$

where A_t is the absolute power of the post-stimulus signal in time window t and R is the absolute power of the baseline signal (-4 to 0 sec.) in a specific frequency band. ERSP for both theta (4-8 Hz) and alpha (8-13 Hz) bands were calculated separately using the Welch method (Welch, 1967). For each trial, the ERSP was calculated for each 5-second interval (*e.g.*, 0-5 sec., 5-10 sec. and so on). The average over all trials for each interval was calculated for each condition.

Different strategies were applied to obtain a more robust estimate of the changes. Firstly, 2 groups of electrodes, representing the frontal lobe and parietal lobe were regrouped : for the frontal lobe, the ERSPs of electrodes AF3, AF4, AF7, AF8, AFz, F1, F2, F3, F4, F5, F6, F7, F8, Fz, FC1, FC2, FC3, FC4, FC5, FC6, and FCz (shown in blue dots in **Fig. 2B**.) and for changes in parietal lobe the ERSPs of electrodes CP1, CP2, CP3, CP4, CPz, P1, P2, P3, P4, P5, P6, P7, P8, Pz, PO3, PO4 and POz (shown in blue dots in **Fig. 2B.**) were averaged together.

Secondly, Cluster-Based Permutation Tests (CBPT) were applied on both Alpha ERSPs and Theta ERSPs. CBPT were used to overcome Multiple Comparison Problem (MCP). This problem arises because of the effect of interest is evaluated at an extremely large number of (sensor, time)-pairs (Maris and Oostenveld, 2007). Applied to the current dataset, CBPT highlighted changes in right frontal lobe for the alpha band between some conditions. Thus, ERSPs of electrodes Fpz, Fp2, AF8, AF4, AFz, Fz, F2, F4, F6 and F8 were averaged together (shown in green dots in **Fig. 2B.**).

Statistical Methods

We used R (R Core Team, 2012) and lme4 package (Bates, Maechler & Bolker, 2012) to perform a linear mixed effects analysis. Linear mixed effects (LME) models were used, with a random intercept for individual to control repeated measures. Seven LME regression models were developed to determine the effect of task parameters on behavioral responses (1 model), MPD (1 model), alpha power (3 models) and theta power (2 models).

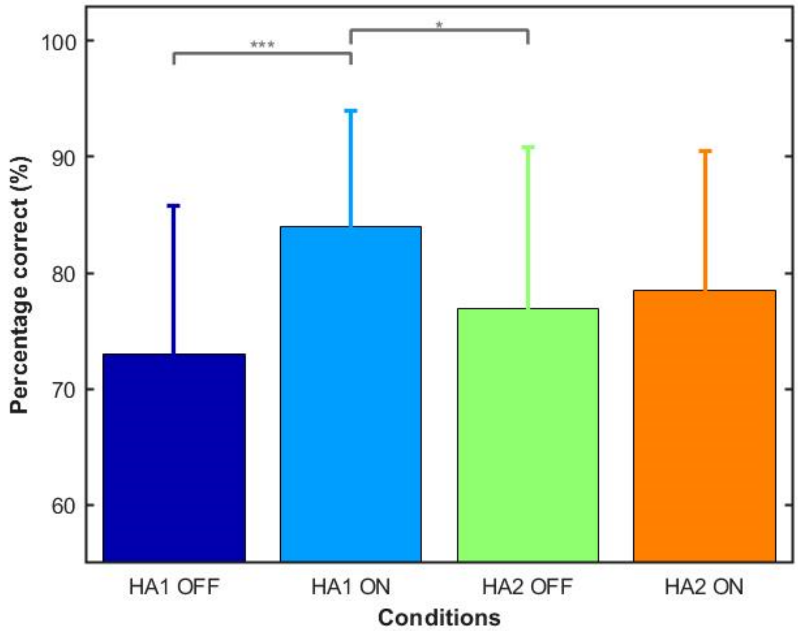

Figure 3. Behavioral results. Each bar represent the mean and the vertical line the standard deviation of percentage correct for the four listening conditions. The significant differences between conditions, extracted from the post hoc analysis, are represented in the form of stars (upper part of the figure). The stars represente intervals of p-value : (*) p-value < 0.05, (**) p-value < 0.01 and (***) p-value < 0.001.

For post-hoc pairwise comparisons reported in the following sections, Tukey's HSD tests were conducted using lsmeans package (Lenth, 2015). Results of the seven regression models and post-hoc analysis are summarized in Annexes 1 to 21.

Results

In this section, results for behavioral responses, pupillometry and EEG are reported. Each figure reports results for the 4 conditions of interest, *i.e.* Hearing Aid 1 with noise reduction switch OFF (**HA1 OFF**), Hearing Aid 1 with noise reduction switch ON (**HA1 ON**), Hearing Aid 2 with noise reduction switch OFF (**HA2 OFF**) and Hearing Aid 2 with noise reduction switch ON (**HA2 ON**).

Behavioral results

Fig. 3. shows the mean and the associated standard deviation of the percentage correct responses for the 2 HA models (HA1 and HA2) and the 2 NR configurations (OFF and ON). Statistical results are presented in Annexes 1 to 3.

The application of a noise reduction induced a significant improvement of correct responses (β = 0.379, *SE* = 0.098, t = 3.879, p < 0.001). Statistical analysis highlights a small effect of gender on the behavioral performance (β = 0.244, *SE* = 0.097, t = 2.503, p = 0.012). Post-hoc analysis performed on the interaction between hearing aid and noise reduction

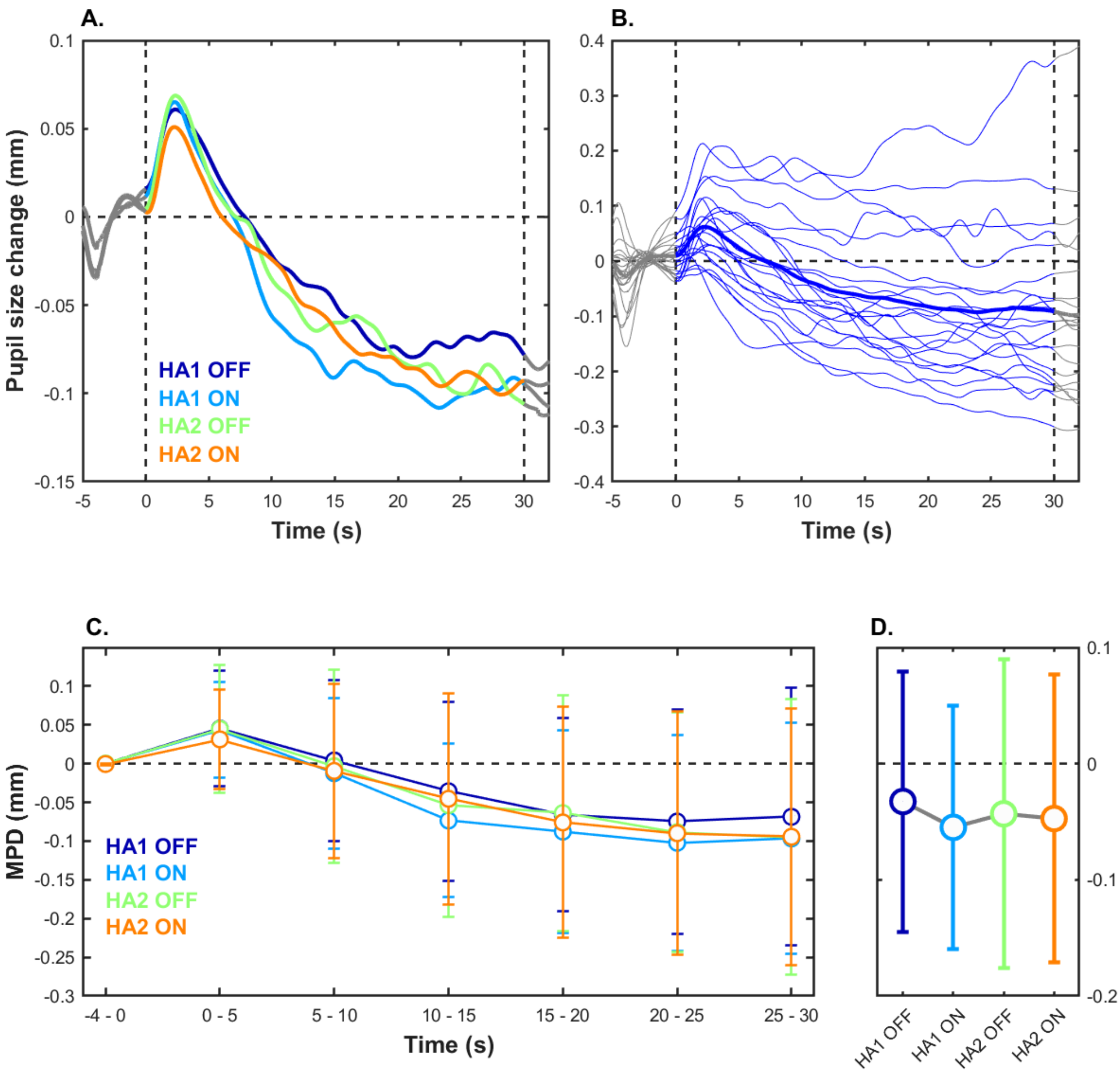

Figure 4. Pupillometry results. A. Pupil size changes relative to baseline in millimeters for the four main conditions. **B.** Individuals pupil size changes relative to baseline in millimeters. **C.** MPD relative to baseline changes in millimeters for the pupillometry data, averaged over each 5-second period. each point represent the mean for a time period, vertical lines represent the standard deviation. **D.** MPD relative to baseline changes in millimeters for the pupillometry data, averaged over the whole trial, *i.e.* 30 seconds. Each point represent the mean for the whole trial, vertical lines the associated standard deviation.

reports two significant differences: between HA1 OFF and HA1 ON and HA1 ON and HA2 OFF.

Pupillometry

To estimate listening effort during the task, the pupil dilation measurement was used as measure of task demand induced by different noise reduction schemes for different hearing aid devices. Results for normalized MPD (*i.e.*, relative to baseline) are compiled in **Fig. 4.** : time courses for the main conditions is shown in the upper left panel, the time

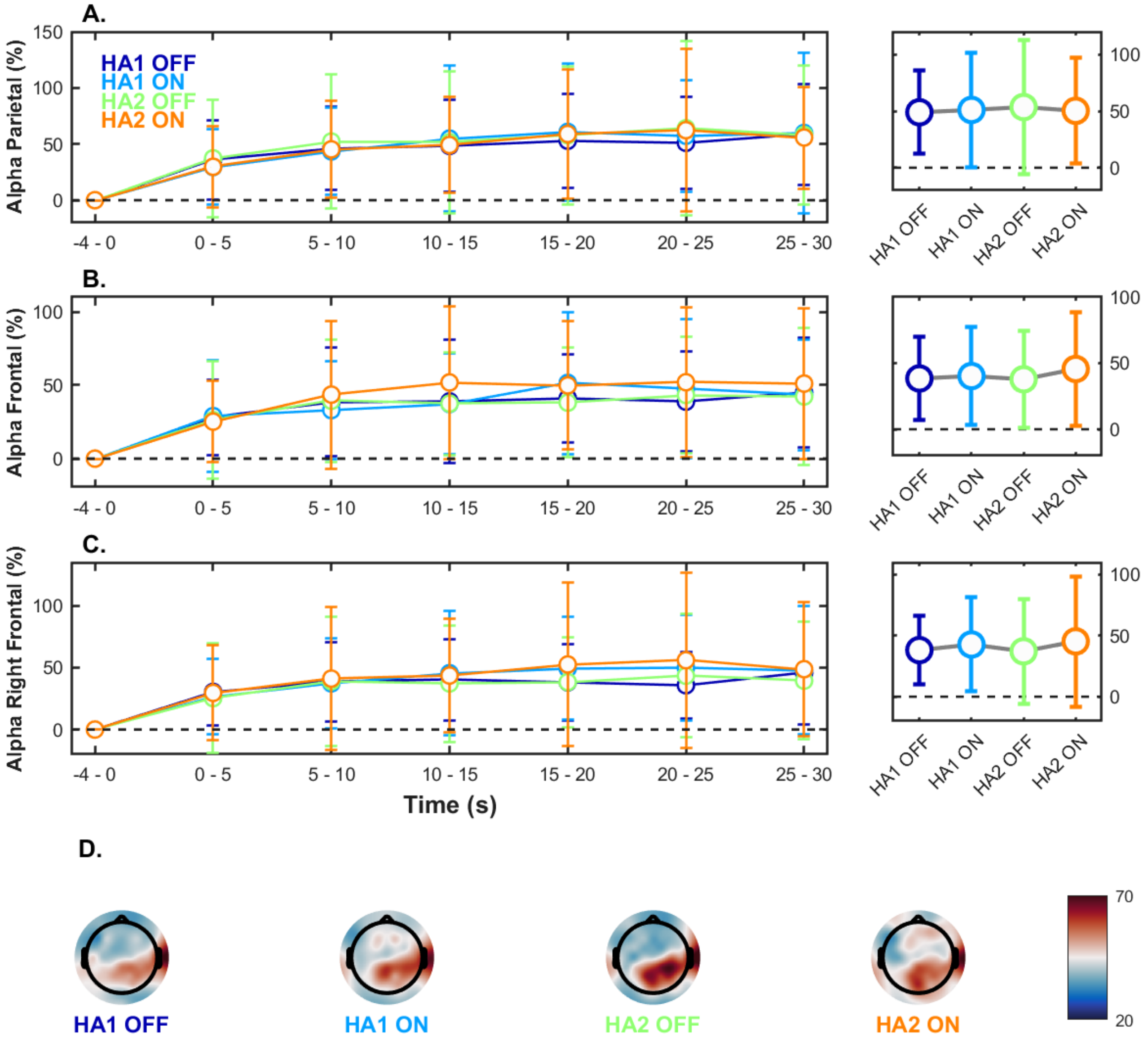

Figure 5. EEG results for Alpha brain wave. A. Left panel : ERSP changes in percentage for the alpha band over the parietal region, averaged over each 5-second period. Right panel: Mean average of parietal alpha over 30 seconds. **B.** Left panel : ERSP changes in percentage for the alpha band over the frontal region, averaged over each 5-second period. Right panel: Mean average of frontal alpha over 30 seconds. **C.** Left panel : ERSP changes in percentage for the alpha band over the right-frontal region, averaged over each 5-second period. Right panel: Mean average of right-frontal alpha over 30 seconds. For each graphic, points represent the mean for the specific time period and vertical lines the associated standard deviation. **D.** Grand average EEG topographical maps over 30 seconds.

courses averaged for each subject are shown in the upper right panel, the time courses averaged in time period (5 seconds) are shown in the lower left panel and the average of normalized MPD over the range of 30 seconds are shown in the lower right panel. Statistical results are presented in Annexes 4 to 6.

For each listening situation, a peak of dilation is observed just after the trial onset (0 to 5 seconds). After that peak, a slight decrease is noticeable during the next 15 seconds (5 to 20 seconds). At the end of trial, a plateau is visible (20 to 30 seconds).

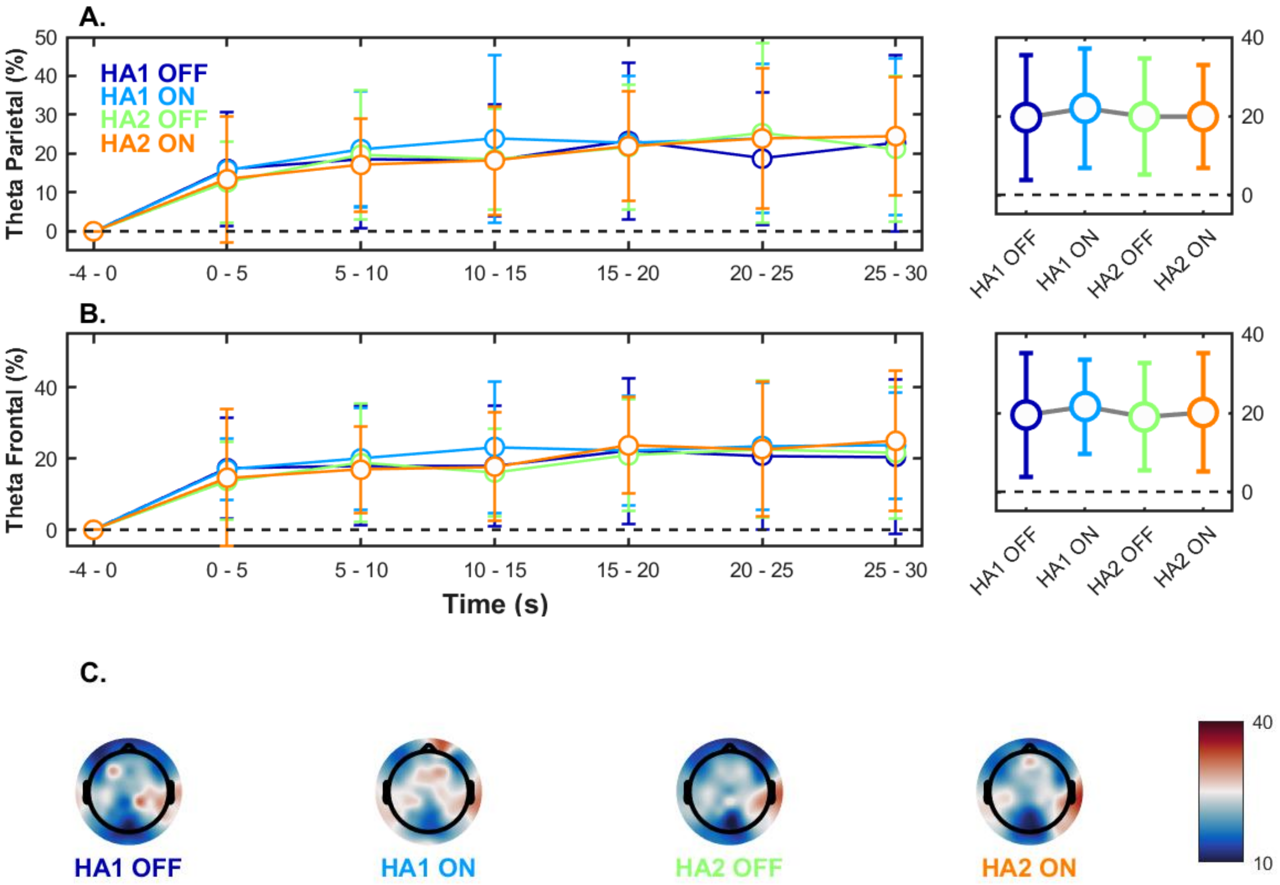

Figure 6. EEG results for Theta brain wave. A. Left panel : ERSP changes in percentage for the theta band over the parietal region, averaged over each 5-second period. Right panel: Mean average of parietal theta over 30 seconds. **B.** Left panel : ERSP changes in percentage for the theta band over the frontal region, averaged over each 5-second period. Right panel: Mean average of frontal theta over 30 seconds. **C.** Grand average EEG topographical maps over 30 seconds.

The comparison between the difference of normalized MPD indicated significantly larger MPD (β = -0.026, *SE* = 0.008, *t* = -3.338, *p* < 0.001) without noise reduction compared to with noise reduction. No significant differences were found between hearing aid models (β = -0.008, *SE* = 0.008, *t* = 1.074, *p* = 0.283) and between the two gender (β = 0.009, *SE* = 0.008, *t* = 1.212, *p* = 0.226). Moreover, interactions between variables were not statically significant.

EEG

Alpha (8-12 Hz) and theta (4-8 Hz) ERSPs were used to estimate the listening effort during the task. ERSPs were averaged by cortical region (parietal, frontal and right frontal for alpha; parietal and frontal for theta). EEG results are divided in two figures: **Fig. 5.** presents results related to alpha and **Fig. 6.** those related to theta. Statistical results are presented in Annexes 7 to 21.

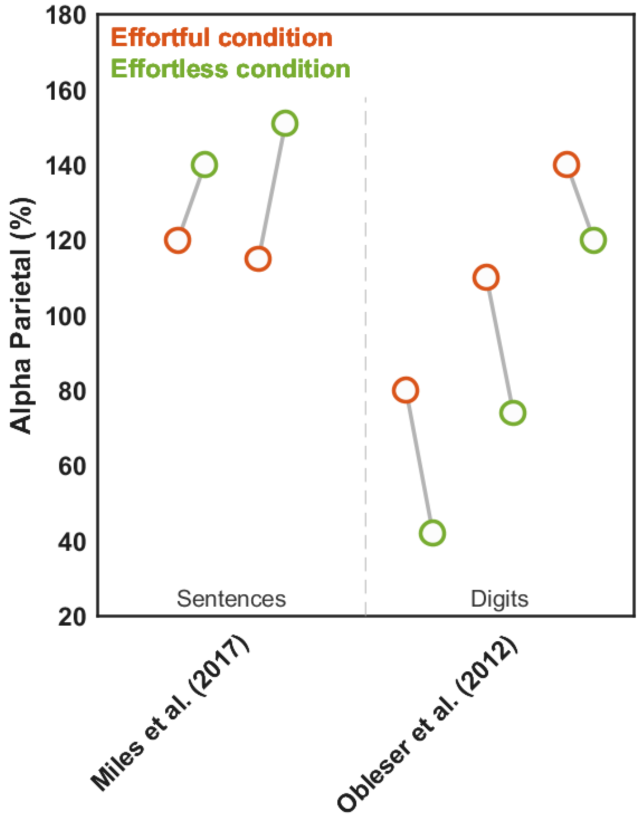

Figure 7. Trends previously observed for alpha power for parietal region. Trends are reported from 2 different studies investigating listening effort. The left panel – “Sentences” illustrated the behavior of alpha power when speech material is “long”. The right panel – “Digits” – illustrated the behavior of alpha power when stimulus is “short”.

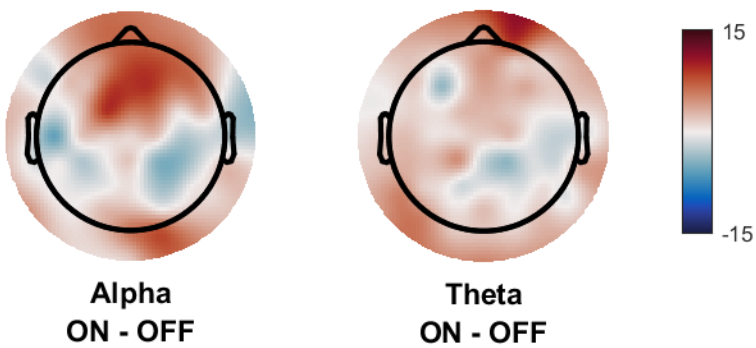

Figure 8. Grand average EEG topographical maps over 30 seconds for differences between the two NR configurations. Left panel: Difference between NR ON and NR OFF for the Alpha power, Right panel: Difference between NR ON and NR OFF for the Theta power. Power is indicated in ERSP (%).

In relation with previous studies which claim that length and complexity of stimulus had an impact on brain waves power, we hypothesized that an increase of alpha and/or theta power was related to a decrease in listening effort.

Knowing this, we could expect more alpha power when NRs were turned on (**H3**). Our results do not indicate significant differences between conditions, regardless of brain waves or cortical region. In some configurations (*e.g.* alpha for frontal) more power seems present when NR is turned ON. However, this trend is statistically significant (alpha frontal: $\beta = 2.518$, $SE = 2.382$, $t = 1.057$, $p = 0.291$; alpha right-frontal: $\beta = 3.909$, $SE = 2.724$, $t = 1.435$, $p = 0.151$).

Discussion

The effect of HA signal processing was investigated with different paradigms in literature: Picou et al. (2013) demonstrated a benefit of signal processing on listening effort in hearing-impaired listeners using a dual-tasks paradigm. Sarampalis et al. (2009) reported a benefit of NR scheme on listening effort using a dual-tasks paradigm. However, in the aforementioned study, the benefit of NR was only demonstrated for participants with normal hearing and at negative SNRs. The review undertaken by Ohlenforst et al. (2017) examined the effect of hearing loss and HA amplification on listening effort. Although some results highlighted a change of cognitive effort (most of them were subjective or behavioral measurements), the authors evoked a lack of evidence of a benefit of HA processing.

In this study, the principal objective was to investigate objectively the impact of noise reduction for different hearing aid models on listening effort during continuous speech. Beyond this first objective, secondary questions have been raised: is continuous speech adapted to assess listening effort? Is +3 dB SNR enough to detect differences?

The speech material, a continuous speech, was chosen to obtain a more ecologically valid approach. Indeed, one major issue arises with traditional hearing testing that measures intelligibility in word or short sentence stimuli: in real life, most listening situations involve conversations with free-running, continuous discourse, not stop after few words, extraction of meaning and generally the formulation of a valid reply (Speaks et al., 1972; MacPherson & Akeroyd, 2013). These additional tasks require more higher-level processing than what is needed to recognize words in isolated sentences (Pichora-Fuller, 2007; Schneider et al., 2010). The choice of a +3 dB SNR is related to previous studies which evaluated environment of hearing aid users. For instance, Smeds et al. (2015) found that hearing aid users are not frequently confronted to negative SNRs or even close to 0 dB. For speech in noise, authors report that average SNR was approximately +5 dB. In the present study, behavioral responses indicate differences with and without noise reduction. That result suggests that listening effort could be also different in the different listening conditions.

Three different assessments were used to evaluate NRs effects: one related to speech performance – Behavioral – and two chosen to reveal changes in the nervous system reflecting listening effort – Pupillometry and EEG.

Behavioral

As hypothesized (**H1**), NRs processing induced an improvement of subject performances. However, this improvement is only significant for one of the two hearing aid models. Noise reduction systems are designed to help users specifically in challenging situations. Acoustically, the objective is the improvement of SNR between relevant and irrelevant signal, through enhancing the speech and/or attenuating the noise (Chung, 2004). The study undertaken by Pearsons et al. (1977) demonstrates that SNRs in daily life conversations are mostly on the order of 5 to 15 dB. In such listening situations, conventional speech recognition tasks may appear insensitive to signal processing improvements

(Rudner et al., 2009; Schulte et al., 2009), and listening effort assessment may be a better predictor of hearing aid signal processing effects.

Listening effort

Pupillometry • Many studies have shown that the pupil response is sensitive to changes in listening effort during presentation of short stimuli: they report that pupil dilation is larger with increasing of listening effort (Zekveld et al., 2018; Ohlenforst et al., 2017; Koelewijn et al., 2012). The use of longer stimuli has resulted of smaller dilation relative to the baseline for each listening situation. More precisely, time courses of pupil size shown a peak in the first 5 seconds, and then a slightly decrease during the 25 seconds remaining. This peak is probably induced by the sensitivity of the pupil to task alertness, which could be more pronounced in the first seconds of the trial. The following relative decrease might be due to evoked pupil dilation to the background noise in the baseline. Similar trends were observed in previous studies using long stimuli (Hjortkjaer et al., 2017; Zhao et al., 2019, Seifi Ala et al., 2020).

In the present study, we hypnotized a diminution of listening effort with NRs, reflected by less dilation of pupil size (**H2**). The MPD (averaged over 30 seconds) was significantly smaller when NRs were activated. Smaller pupil dilation has been associated with increased workload and greater allocation of resources to perform the listening task (Wendt et al., 2017). Wendt et al. (2017) investigated the impact of NR schemes by using Danish sentences from the Hearing In Noise Test (HINT), assimilated to "short" stimuli (1.5 seconds per sentences). Pupil dilation was reduced when NR scheme was applied, suggested a reduced processing. To our knowledge, the present study is the first to show the benefit of NR scheme on processing effort by using long continuous speech.

EEG • Our results do not show a significant increasement neither in alpha power nor in Theta Power when the NR is turned ON. Thus, our hypothesis (**H3**) is not confirmed. However, if we pay attention to differences between ON and OFF conditions (**Fig. 8.**), we globally see that the increase of power is more marked in ON condition. If we focus on alpha power, this observation is consistent with the inhibition hypothesis, where activity that is related to goal-directed task is actively inhibited (Klimesch et al., 2007). However, alpha power as an outcome measure of listening effort can lead to contradictory results. Some studies suggest that the relationship between task difficulty and alpha power is direct. In other words, more difficulty induces increased alpha. This trend has been reported in numerous studies (Obleser et al., 2012; Beckeret al., 2013; Wöstmann et al., 2015; Wisniewski et al., 2017; Wöstmann et al., 2017; Seifi Ala et al., 2020). Other studies report that alpha power decreased when task difficulty increased (Petersen et al., 2015, Miles et al, 2017; Hjortkjaer et al., 2017).

The contradictory of the findings is illustrated in **Fig. 7.**. For example, Seifi Ala et al. (2020) reported a greater alpha power for the parietal region at high SNR (easier condition) in comparison to low SNR (harder condition). The speech target (continuous speech) and irrelevant noise (four talker babble noise and unattended speaker) used are the same in the current study. McMahon et al. (2016), which used degraded sentences with a four-talker babble noise background found the same trend. However, other studies showed opposite results, with a decreased of alpha power in listening effortless situation. In these studies, less complex linguistic and shorter stimuli were used (digit task: Obleser et al., 2012; Wöstmann et al., 2015; word comprehension: Becker, Pefkou, Michel, & Hervais-Adelman, 2013; Obleser & Weisz, 2012).

These opposites findings could suggest that the alpha network is differently influenced by the linguistic complexity and the length of the speech stimuli. Two theories tried to explain these contradictories results : on one hand, the inhibition theory explains that increased alpha is a sign of suppression of unattended sound sources and inhibition of task-irrelevant cortical regions, which consequently should increase alpha with increased difficulty. On the other hand, cortical idling theory. Introduced by Adrian and Matthews (1934), this theory claims that the enhancement of alpha band be a correlate of a deactivated cortical network (Pfurtscheller, 2000). It is assumed that increased alpha power in a cortical region likely reflects decreased neural activation by suppression non-crucial or potentially distracting information allowing for more efficient neural processing. During continuous speech and when noise reduction is activated, due to attenuation of irrelevant content, there may be less dependency on the semantic context to recognize a sentence; whereas without noise reduction, there is a greatest need to rely on semantic context to “fill in the gaps”. However, before semantic processing is engaged, at least some phonemes must be recognized in the incoming speech signal. This lower level phonemic processing may be more demanding without noise reduction, and potentially affects the possibly for mode automated semantic recognition. Increase of alpha power with noise reduction may therefore reflect task-irrelevant inhibition due to the automaticity of speech processing, while reduced alpha power without noise reduction reflect ongoing active processing.

Conclusion

In this exploratory study pupillometry and EEG were used to assess aspects of listening effort of hearing-aid users in a continuous speech setting. When listening to 30-second news clips, presented from either a right or left target in the presence of 4-talker babble noise, better comprehension was observed with NR processing, lower listening effort was observed with pupillometry for the less demanding and effortful condition (with NR), and EEG analysis do not report significant effect of NRs on alpha and theta, even if a global increasement of alpha and theta brain waves was found.

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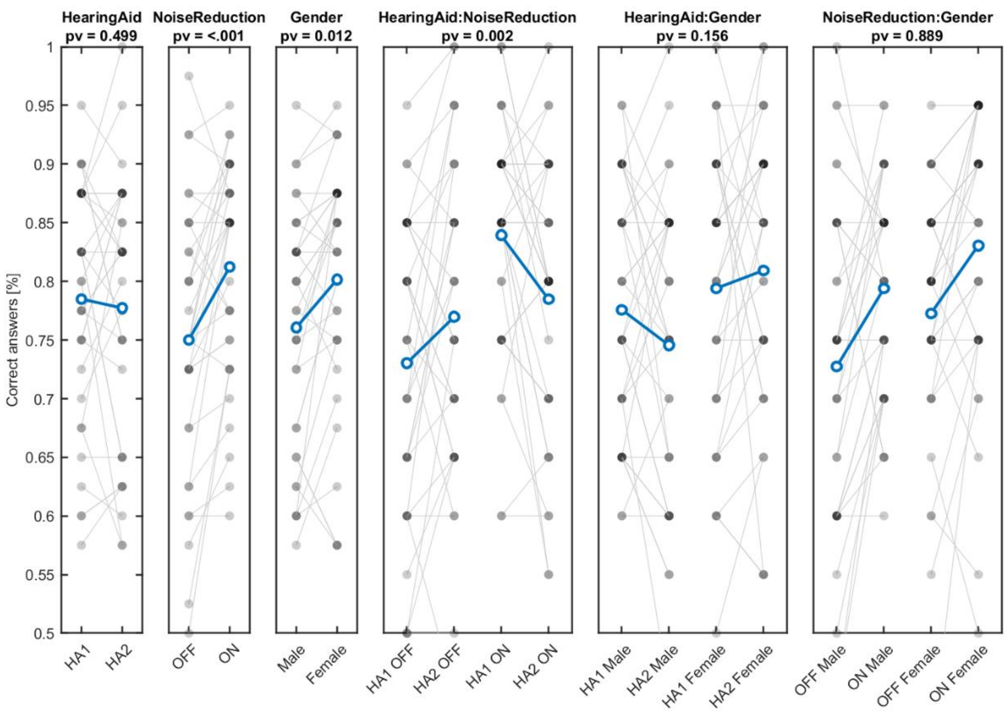
Annexes

	Correct response		
	Estimate	95% CI	p-value
Intercept	1.423	[1.165, 1.682]	<.001
Hearing Aid	-0.066	[-0.257, 0.125]	0.499
Noise Reduction	0.379	[0.187, 0.571]	<.001
Talker	0.244	[0.053, 0.435]	0.012
Hearing Aid : Noise Reduction	-0.59	[-0.973, -0.208]	0.002
Hearing Aid : Talker	0.277	[-0.105, 0.659]	0.156
Noise Reduction : Talker	-0.027	[-0.409, 0.355]	0.355

Annexe 1. Linear Mixed Model for behavioral responses.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	-0.23	0.132	0.3
HA1 OFF	HA1 ON	-0.693	0.141	<0.001
HA1 OFF	HA2 ON	-0.326	0.134	0.070
HA2 OFF	HA1 ON	-0.463	0.145	0.008
HA2 OFF	HA2 ON	-0.096	0.137	0.898
HA1 ON	HA2 ON	0.367	0.146	0.059

Annexe 2. Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for behavioral responses.



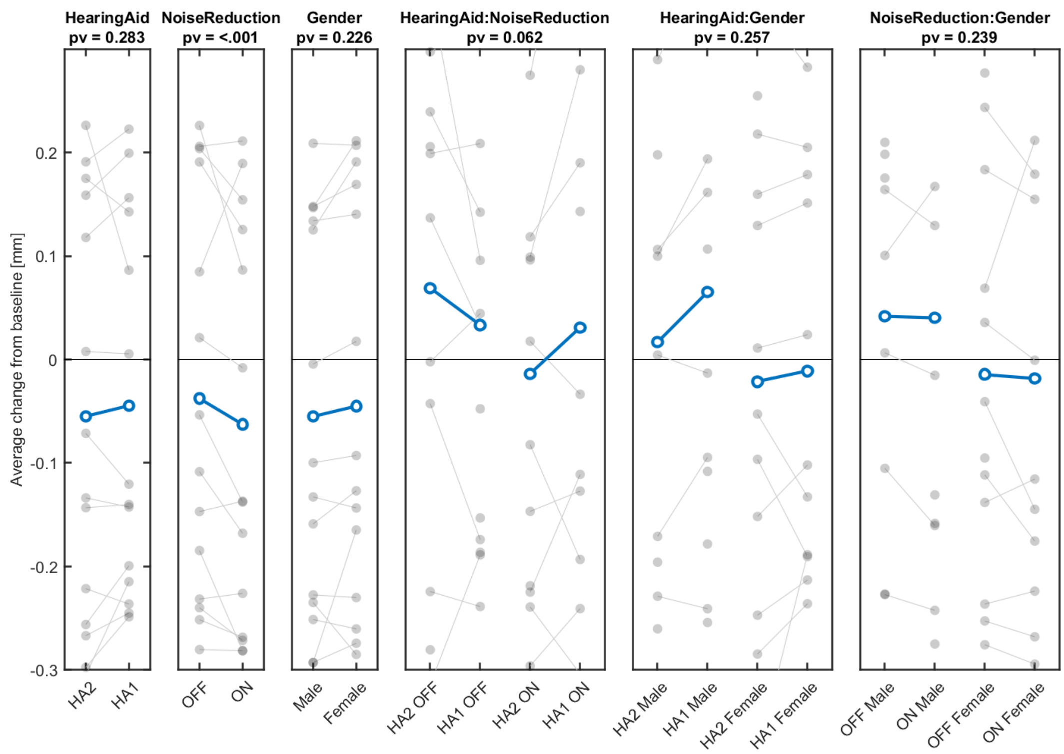
Annexe 3. Behavioral results extracted from the statistical model. The three plots placed at left reprensent main effects, the three plots placed at right reprensents interactions.

	Mean Pupil Dilation		
	Estimate	95% CI	p-value
Intercept	-0.05	[-0.185, 0.085]	0.468
Hearing Aid	0.008	[-0.007, 0.023]	0.283
Noise Reduction	-0.026	[-0.041, -0.106]	<.001
Gender	0.009	[-0.006, 0.024]	0.226
Hearing Aid : Noise Reduction	0.028	[-0.001, 0.058]	0.06
Hearing Aid : Gender	0.017	[-0.013, 0.047]	0.257
Noise Reduction : Gender	0.018	[-0.012, 0.048]	0.239

Annexe 4. Linear Mixed Model for the Mean Pupil Dilation.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	0.006	0.011	0.944
HA1 OFF	HA1 ON	0.04	0.011	0.001
HA1 OFF	HA2 ON	0.017	0.011	0.394
HA2 OFF	HA1 ON	0.034	0.011	0.008
HA2 OFF	HA2 ON	-0.096	0.011	0.717
HA1 ON	HA2 ON	-0.022	0.011	0.167

Annexe 5. Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for the Mean Pupil Dilation.



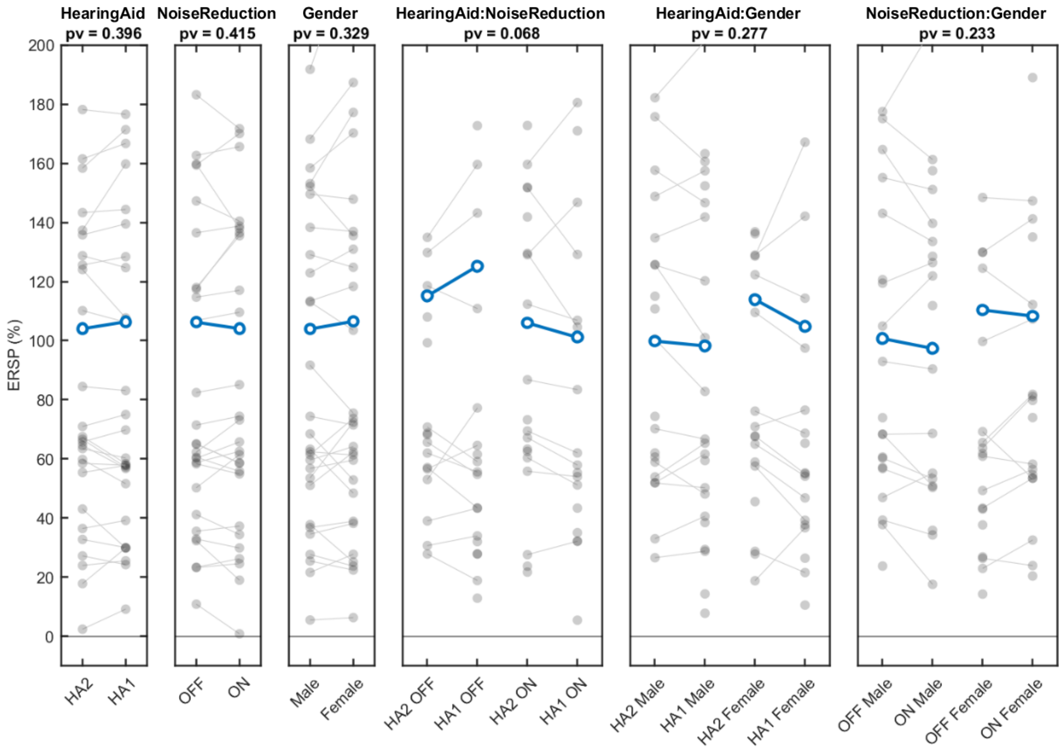
Annexe 6. Mean Pupil Dilation results extracted from the statistical model. The three plots placed at left reprensents main effects, the three plots placed at right reprensents interactions.

	Alpha Parietal		
	Estimate	95% CI	p-value
Intercept	105.19	[74.56, 135.8]	<.0001
Hearing Aid	2.222	[-2.906, 7.351]	0.396
Noise Reduction	-2.131	[-7.26, 2.997]	0.415
Gender	2.552	[-2.578, 7.682]	0.329
Hearing Aid : Noise Reduction	-9.553	[-19.82, 0.713]	0.068
Hearing Aid : Gender	-5.687	[-15.95, 4.573]	0.277
Noise Reduction : Gender	6.24	[-4.018, 16.5]	0.233

Annexe 7. Linear Mixed Model for the Alpha Parietal Power.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	-6.999	3.71	0.233
HA1 OFF	HA1 ON	-2.645	3.71	0.892
HA1 OFF	HA2 ON	-0.091	3.71	1
HA2 OFF	HA1 ON	4.353	3.7	0.642
HA2 OFF	HA2 ON	6.908	3.7	0.244
HA1 ON	HA2 ON	2.554	3.71	0.901

Annexe 8. Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for the Alpha Parietal Power.



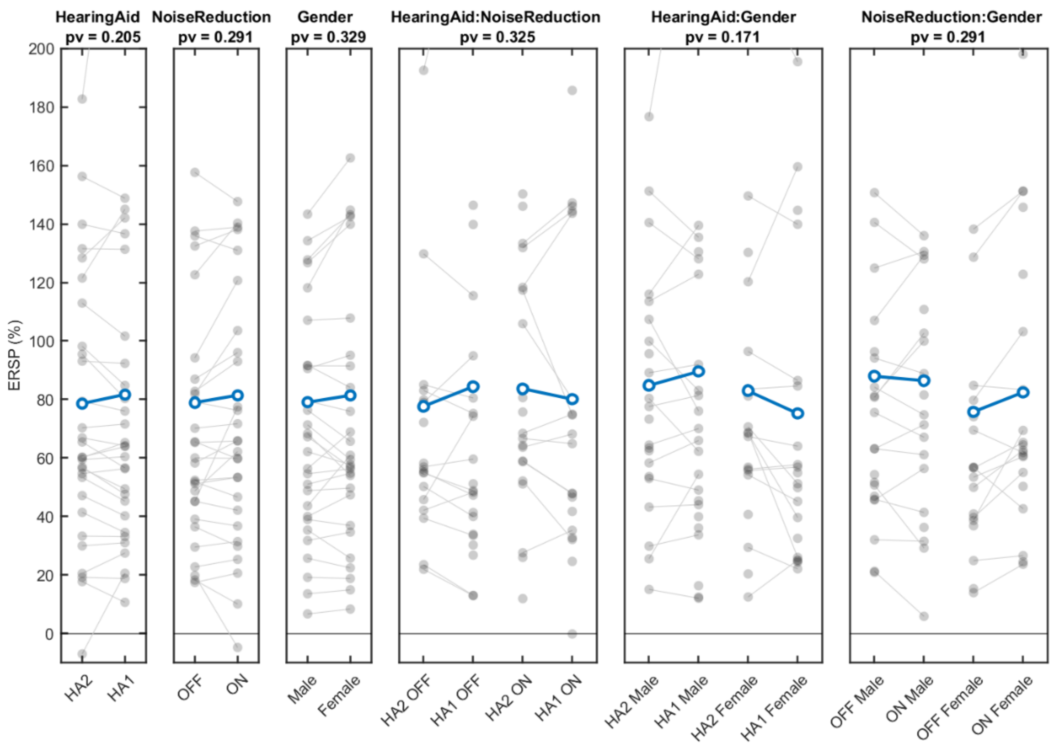
Annexe 9. Results for the Alpha Parietal Power extracted from the statistical model. The three plots placed at left reprensents main effects, the three plots placed at right reprensents interactions.

	Alpha Frontal		
	Estimate	95% CI	p-value
Intercept	80.20	[59.86, 100.5]	<.0001
Hearing Aid	3.023	[-1.648, 7.694]	0.204
Noise Reduction	2.518	[-2.152, 7.189]	0.29
Gender	2.323	[-2.345, 6.992]	0.329
Hearing Aid : Noise Reduction	-4.698	[-14.05, 4.655]	0.325
Hearing Aid : Gender	-6.529	[-15.87, 2.81]	0.17
Noise Reduction : Gender	5.03	[-4.304, 14.37]	0.29

Annexe 10. Linear Mixed Model for for the Alpha Frontal Power.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	-5.372	3.38	0.385
HA1 OFF	HA1 ON	-4.867	3.38	0.474
HA1 OFF	HA2 ON	-5.541	3.38	0.356
HA2 OFF	HA1 ON	0.505	3.37	0.999
HA2 OFF	HA2 ON	-0.169	3.38	1
HA1 ON	HA2 ON	-0.674	3.37	0.997

Annexe 11 Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for the Alpha Frontal Power.



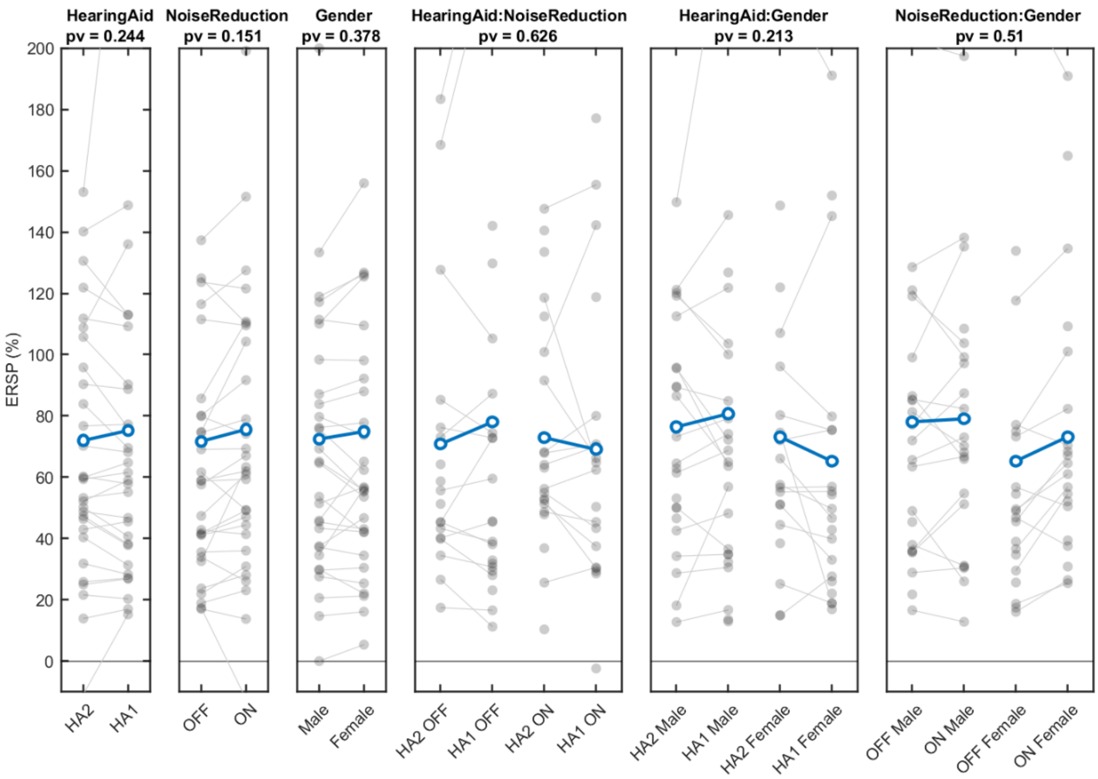
Annexe 12. Results for the Alpha Parietal Power extracted from the statistical model. The three plots placed at left reprensents main effects, the three plots placed at right reprensents interactions.

	Alpha Right-Frontal		
	Estimate	95% CI	p-value
Intercept	73.66	[54.73, 92.6]	<.0001
Hearing Aid	3.177	[-2.165, 8.519]	0.243
Noise Reduction	3.909	[-1.432, 9.25]	0.151
Gender	2.399	[-2.938, 7.736]	0.378
Hearing Aid : Noise Reduction	-2.655	[-13.35, 8.038]	0.626
Hearing Aid : Gender	-6.779	[-17.46, 3.899]	0.213
Noise Reduction : Gender	3.59	[-7.086, 14.27]	0.51

Annexe 13. Linear Mixed Model for for the Alpha Right-Frontal Power.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	-4.505	3.86	0,649
HA1 OFF	HA1 ON	-5.236	3.86	0.527
HA1 OFF	HA2 ON	-7.086	3.86	0.257
HA2 OFF	HA1 ON	-0.731	3.86	0.998
HA2 OFF	HA2 ON	-2.581	3.86	0.909
HA1 ON	HA2 ON	-1.85	3.86	0.964

Annexe 14. Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for the Alpha Right-Frontal Power.



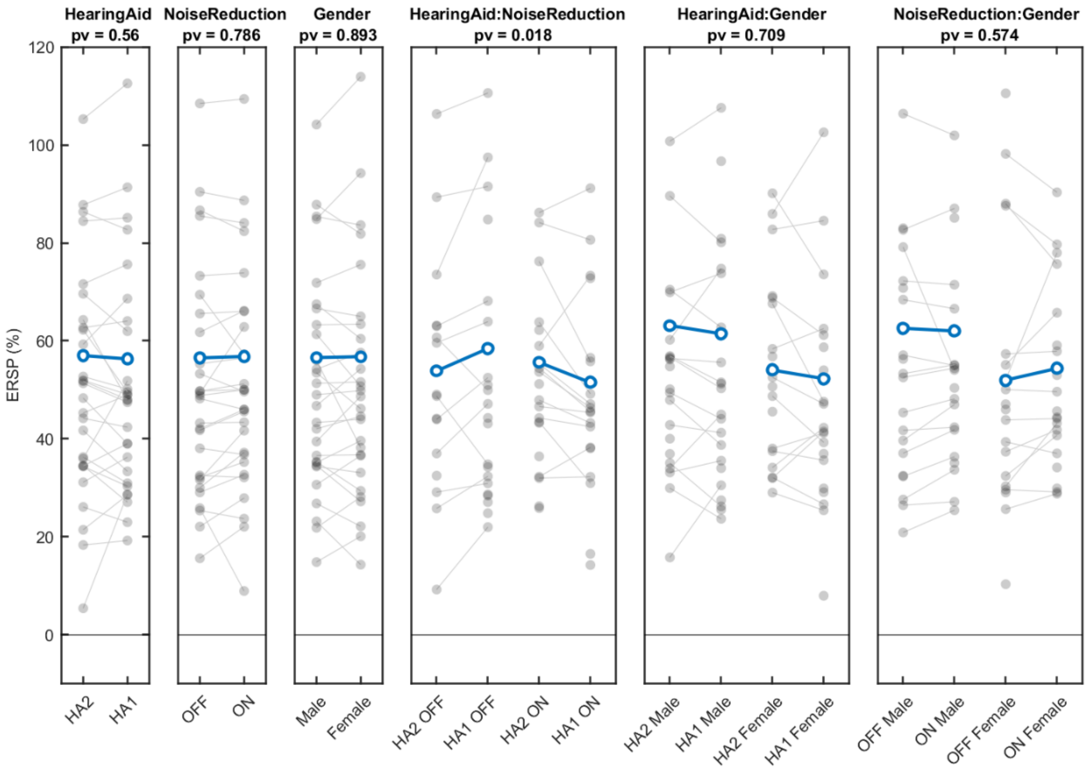
Annexe 15. Results for the Alpha Right-Frontal Power extracted from the statistical model. The three plots placed at left reprensents main effects, the three plots placed at right reprensents interactions.

	Theta Parietal		
	Estimate	95% CI	p-value
Intercept	56.67	[45.06, 68.28]	0
Hearing Aid	-0.683	[-2.985, 1.618]	0.561
Noise Reduction	0.318	[-1.983, 2.62]	0.786
Gender	0.158	[-2.143, 2.46]	0.893
Hearing Aid : Noise Reduction	-5.57	[-10.18, -0.96]	0.018
Hearing Aid : Gender	0.877	[-3.727, 5.481]	0.709
Noise Reduction : Gender	1.318	[-3.284, 5.921]	0.574

Annexe 16. Linear Mixed Model for the Theta Parietal Power.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	-2.101	1.67	0.588
HA1 OFF	HA1 ON	-3.103	1.66	0.244
HA1 OFF	HA2 ON	0.365	1.66	0.996
HA2 OFF	HA1 ON	-1.002	1.66	0.931
HA2 OFF	HA2 ON	2.466	1.66	0.448
HA1 ON	HA2 ON	3.468	1.66	0.16

Annexe 17. Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for the Theta Parietal Power.



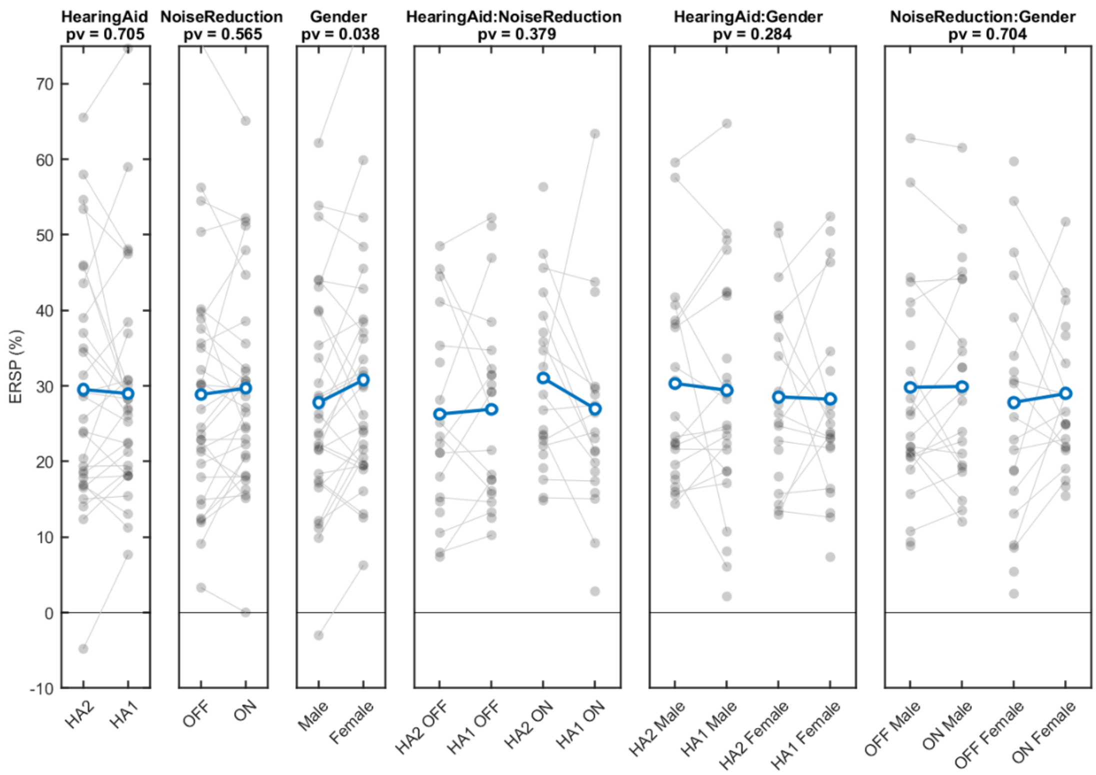
Annexe 18. Results for the Theta Parietal Power extracted from the statistical model. The three plots placed at left reprensents main effects, the three plots placed at right reprensents interactions.

	Theta Frontal		
	Estimate	95% CI	p-value
Intercept	29.27	[23.53, 35]	0
Hearing Aid	-0.537	[-3.315, 2.241]	0.705
Noise Reduction	0.815	[-1.964, 3.594]	0.565
Gender	2.937	[0.16, 5.715]	0.038
Hearing Aid : Noise Reduction	-2.495	[-8.056, 3.066]	0.379
Hearing Aid : Gender	3.035	[-2.521, 8.592]	0.284
Noise Reduction : Gender	1.077	[-4.479, 6.633]	0.704

Annexe 19. Linear Mixed Model for the Theta Frontal Power.

Contrast		Estimate	SE	p-value
HA1 OFF	HA2 OFF	-0.71	2.01	0.985
HA1 OFF	HA1 ON	-2.063	2.01	0.734
HA1 OFF	HA2 ON	-0.278	2.01	0.999
HA2 OFF	HA1 ON	-1.353	2.01	0.907
HA2 OFF	HA2 ON	0.432	2.01	0.996
HA1 ON	HA2 ON	1.785	2.01	0.81

Annexe 20. Post-hoc analysis for the interaction between Hearing Aid and Noise Reduction for the Theta Frontal Power.



Annexe 21. Results for the Theta Frontal Power extracted from the statistical model. The three plots placed at left reprensents main effects, the three plots placed at right reprensents interactions.